Calculating Kinetic Energy Worksheet

1. Kinetic energy can be defined as…..

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2. The equations to be used to calculate kinetic energy and speed are….

3. The SI (standard) unit in which kinetic energy is often measured is the _____

4. Michael Clarke swung a bat which had a mass of 2 kg at a velocity of 45 m/s. How many joules of kinetic energy could he give to a ball?

5. Barry Bonds swings a bat which has a mass of 1.5 kg at a velocity of 55 m/s. How many joules of kinetic energy could he give to a ball?

6. A golf pro has 2100 J of kinetic energy and swings his driver which weighs .75 kg. What is the speed of his swing?

7. A soccer ball has 1200 J of kinetic energy and a mass of 0.8 kg. What is the speed of the ball when it is kicked?

8. An AFL player has 3500 J of kinetic energy and is running at a speed of 8 m/s. Calculate his mass.

9. Explain the following as fully as you can:

   a) In school zones the speed limit is reduced to 40km/hr (just over 10m/s) during school times.

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   b) A car travelling at 14m/s would cause less damage when it crashed into a wall, than a large truck travelling at the same velocity?

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